

Physical Chemistry (I)

Lecture 25



기말고사 안내

- 시간: 6월 17일(목) 오후 1:00~2:30
- 장소: 화학관 315호(교수계획표에 나온 수업 장소)
- 계산기 불필요

시험 범위

- 7장 “단순 혼합물” (2차 중간고사 범위 이후)
- 8장 “화학 평형”
- 9장 “전기화학”
- 10장 “통계역학”

시험 문제

- 총 여덟 문제
- 과제 문제 혹은 그 변형
- 수업 시간에 배운 개념 설명
- 수업 시간에 배운 공식 유도
- 응용 문제

Chapter 7

- Phase
- Chemical potential
- Gibbs-Duhem eq.



Multi-component
system



Ternary systems



Idealized system

Ideal solution

- Raoult's law

$$p_A = x_A p_A^*$$

- Chemical potential

$$\mu_A = \mu_A^* + RT \ln x_A$$

- $\Delta_{\text{mix}} G = -T \Delta_{\text{mix}} S$

Ideal-dilute solution

- Henry's law

$$p_B = x_B K_B$$

- Chemical potential

$$\mu_B = \mu_B^\ominus + RT \ln x_B$$

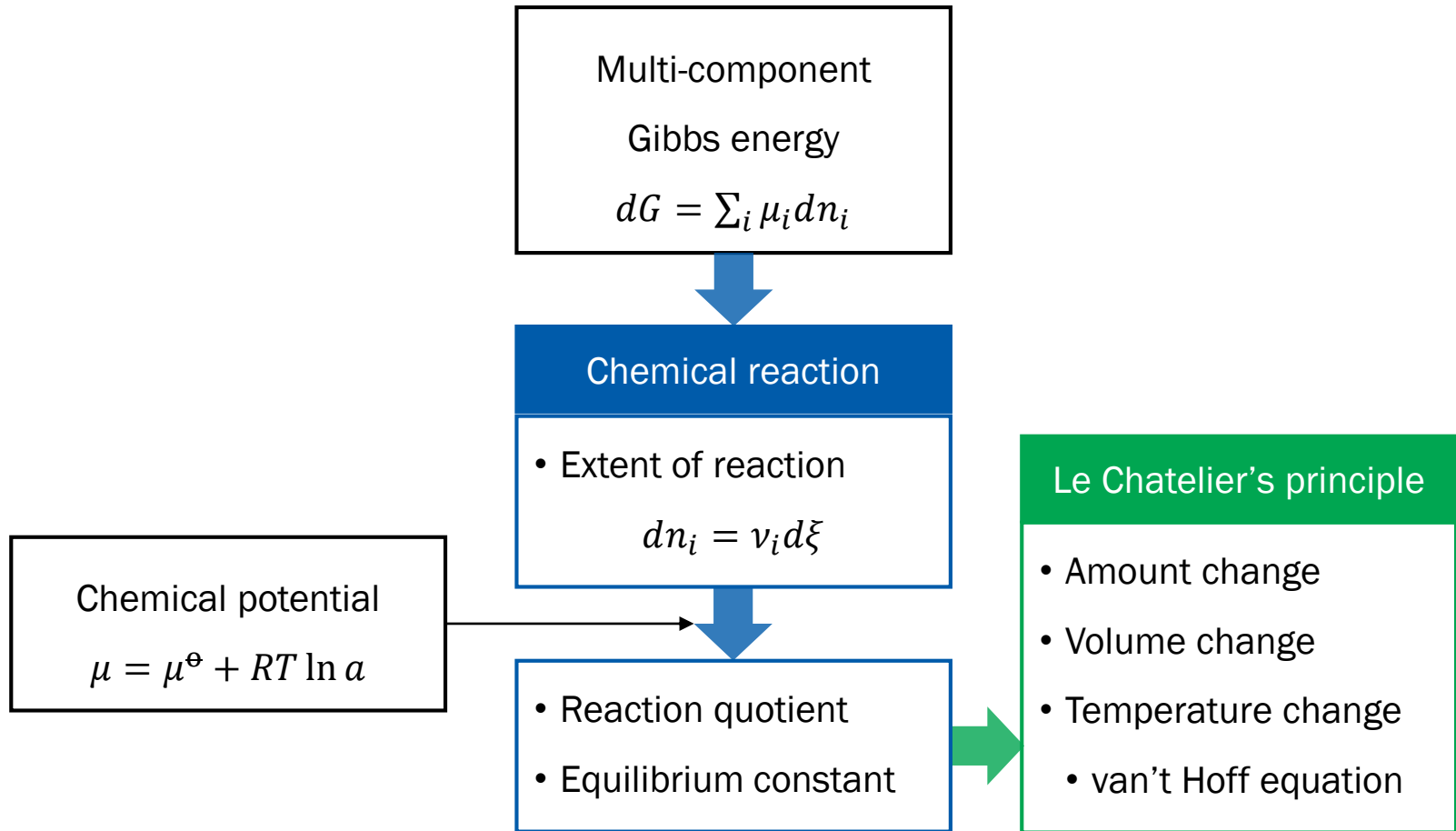
- L-G phase diagram
- L-L phase diagram
- phase separation
- L-S phase diagram

Colligative properties

Concepts and Derivations in Chpt 7

- [C] Phase diagrams of real solutions
 - [C] Distillation
 - [C] Liquid-gas equilibria: azeotropes
 - [C, D] Liquid-liquid equilibria: phase separation
- [C] Phase diagrams of liquid-solid mixtures
- [C, D] Colligative properties
 - Lowering of vapor pressure
 - Elevation of T_b , depression of T_f , solubility
 - Osmotic pressure
- [C] Phase diagrams of ternary systems

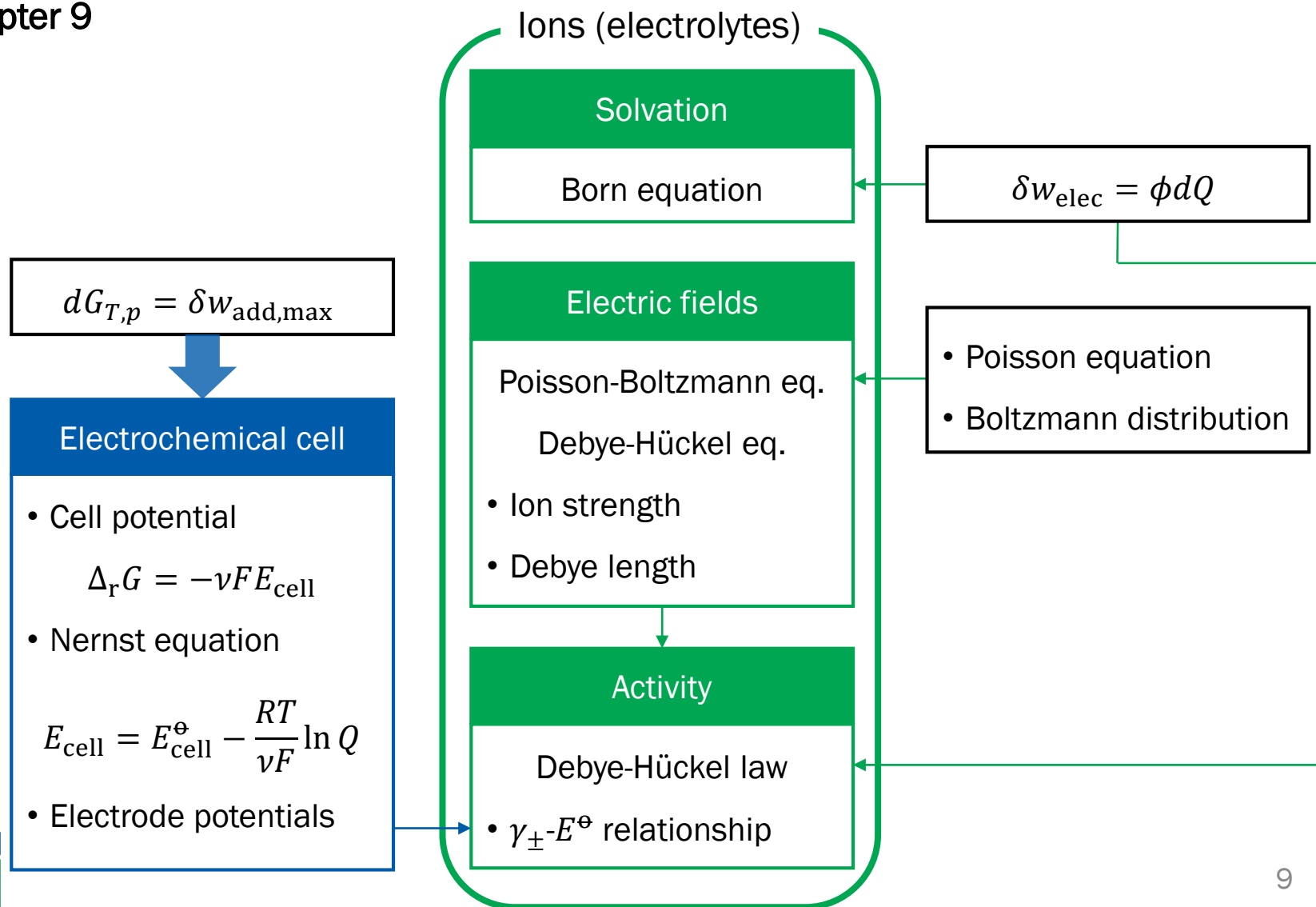
Chapter 8



Concepts and Derivations in Chpt 8

- [C] Chemical potential and activity
- [C, D] Reaction quotient and equilibrium constant
- [D] Activity of liquids and solids
- [D] Gas reactions
- [C, D] Le Chatelier's principle
 - Amount change
 - Volume change
 - Temperature change: van't Hoff equation

Chapter 9

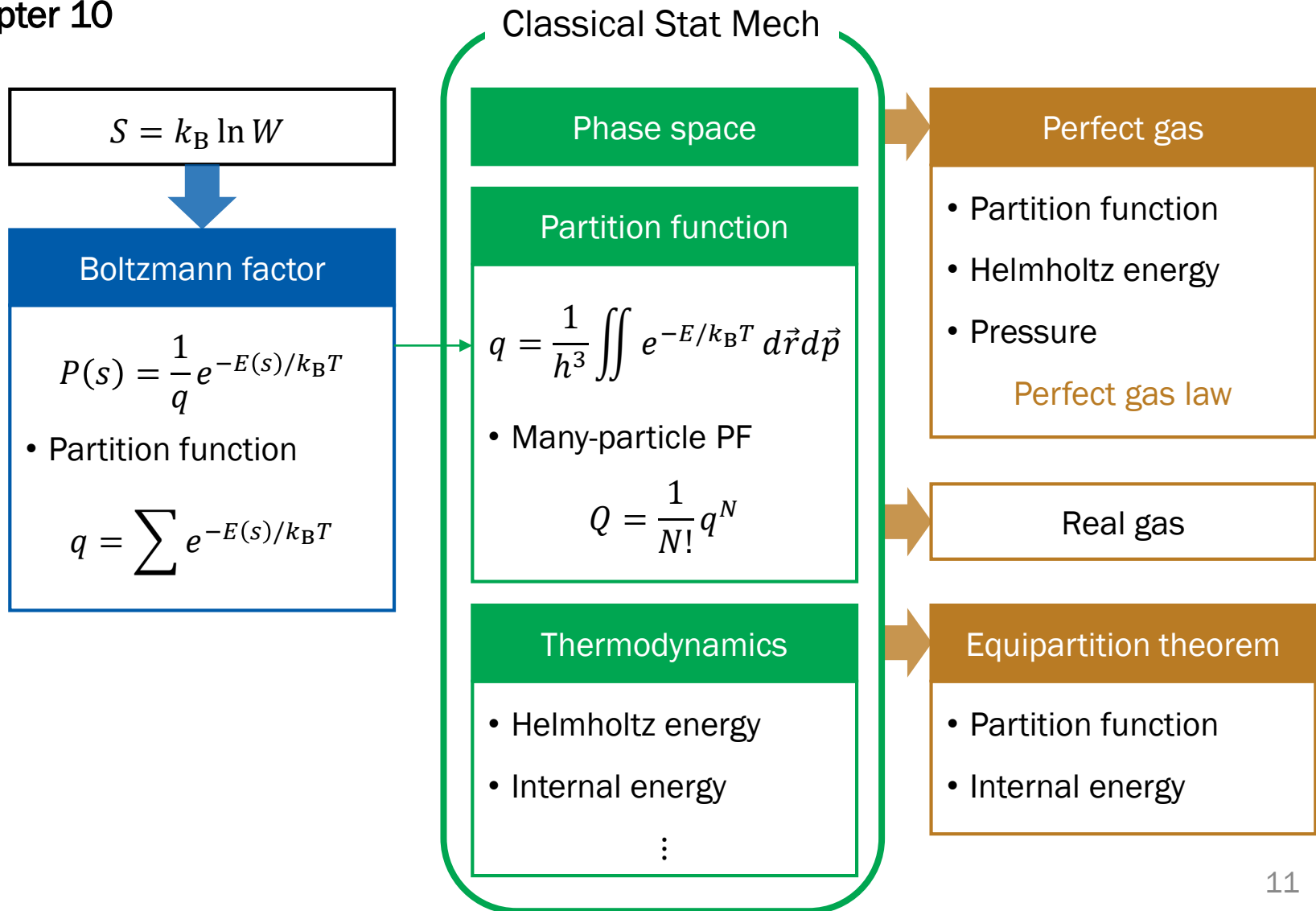


Concepts and Derivations in Chpt 9

- [C] Electrochemical cells
- [D] Thermodynamics of cells
- [D] Nernst equation
- [D] Entropy and enthalpy of cell reactions
- [C, D] Electrode potentials
- [C, D] Gibbs energy of solvation: Born equation
- [D] Ion potential: Debye-Hückel equation
- [D] Ion activity: Debye-Hückel limiting law



Chapter 10



Concepts and Derivations in Chpt 10

- [D] Boltzmann factor
- [C] Partition function
- [C] Phase space
- [C] Partition function of classical system
 - [D] 1-dimensional spring
- [C] Indistinguishability

Concepts and Derivations in Chpt 10

- [D] Partition function and thermodynamics
 - Internal energy
 - Helmholtz energy
 - Other thermodynamic quantities
- [D] Perfect gas: perfect gas law
- [C] Real gas: van der Waals equation
- [D] Equipartition theorem